# Technical Guide: Implementing AspeQt-2k26 on Raspberry Pi 5

## Subtitle: High-Speed SIO R: Emulation via Native GPIO and Qt6

**Author:** 13leader.net

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**Project Repo:** https://github.com/pjones1063/AspeQt-2k26

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## 1. Overview

As modern operating systems deprecate the Qt5 libraries, maintaining stable SIO (Serial Input/Output) connectivity for the Atari 8-bit family requires a transition to Qt6. AspeQt-2k26 provides this future-proofing while introducing native TNFS browsing and advanced virtual modem capabilities. This document outlines the steps to install the portable 64-bit build on a Raspberry Pi and interface it directly with Atari hardware via the GPIO header.

**Note on USB Adapters:** While this guide focuses specifically on utilizing the Raspberry Pi's native GPIO header for microsecond-accurate R: Device emulation, recent updates to AspeQt-2k26 have also unlocked this feature for standard USB FTDI adapters (by addressing legacy RS-232 logic inversion). However, the direct GPIO method outlined here remains the most robust, latency-free option for dedicated Pi setups.

## 2. Installation (ARM64 Portable)

Instead of utilizing standard repository installs which may carry legacy dependencies, use the optimized portable tarball. You must also install the libgpiod C++ v2 bindings, which are strictly required for the R: Device's hardware-level SIO COMMAND line interrupts.

## Download:

```
$ sudo apt update
```

```
$ sudo apt install qt6-base-dev qt6-base-private-dev libqt6gui6
libqt6widgets6 libqt6network6 libgpiod-dev libgpiodcxx-dev
```

`$ wget https://github.com/pjones1063/AspeQt-2k26/release` *(add current release – Pi tarball)*

### Extract:

```
$ tar -xzvf AspeQt-2K26-pi.64-portable.tar.gz
```

### Execute:

```
$ cd AspeQt-2K26-pi.64-portable $ ./AspeQt
```

## 3. Hardware Interfacing (GPIO to SIO)

The Raspberry Pi 4/5 can communicate directly with the Atari SIO bus using its internal UART.

[CRITICAL SAFETY NOTE] Atari SIO logic levels are 5V. The Raspberry Pi GPIO header operates at 3.3V. Connecting these directly will result in permanent damage to the Pi's SoC. A Logic Level Shifter is mandatory for this project.

Wiring Diagram Reference (the 3.3V vs 5V Requirement) The following table outlines the required connections. *Note: Wiring the COMMAND line (SIO Pin 7 to Pi Pin 12) is absolutely mandatory if you intend to use the R: Device 850 Emulation.*

**Wiring Diagram Reference (the 3.3V vs 5V Requirement) Logic Shifter Wiring Table**

The following table:

| **Connection Type** | **Atari SIO (5V Side)** | **Shifter Channel** | **Pi GPIO (3.3V Side)** |
|---|---|---|---|
| Data In (Atari RX) | SIO Pin 3 **[Diode]** | HV1 <---> LV1 | Pin 8 (GPIO 14 / TX) |
| Data Out (Atari TX) | SIO Pin 5 | HV2 <---> LV2 | Pin 10 (GPIO 15 / RX) |
| Command (Atari CMD) | SIO Pin 7 | HV3 <---> LV3 | Pin 12 (GPIO 18) |
| Reference Power | SIO Pin 10 (+5V) | HV <---> LV | Pi Pin 1 (+3.3V) |
| Common Ground | SIO Pin 4 (GND) | GND <---> GND | Pi Pin 6 (GND) |

**[CRITICAL HARDWARE REQUIREMENT: The Bus Contention Diode]**

Because the Atari SIO bus is a shared daisy-chain, modern push-pull logic shifters will forcefully hold the SIO data line at 5V, overpowering vintage hardware (like the 1050 disk drive or 850 interface) and causing short circuits. To safely emulate an open-collector output and protect your vintage hardware, you must install a switching diode (1N4148 or 1N914) inline on the Pi's Transmit TX line (Pin 8 GPIO 14).

- Location: Between the Logic Shifter's HV1 output and Atari SIO Pin 3.
- Orientation: The diode's black stripe (Cathode) must point towards the Raspberry Pi's logic shifter. The non-stripe side (Anode) connects to the Atari.
- The Raspberry Pi GPIO header operates at 3.3V, while the Atari SIO bus operates at 5V. Connecting these directly will cause permanent hardware failure on the Raspberry Pi 4/5.

**4. System Configuration**

Before AspeQt can access the serial hardware, the Pi's default "Serial Console" must be disabled to release /dev/serial0. Open the configuration tool: sudo raspi-config Navigate to Interface Options -> Serial Port. Set *Login shell over serial* to NO. Set *Hardware serial port* to YES. Finish and Reboot.

**5. Software Configuration:**

Disk Drives & SIO Launch AspeQt-2k26 and enter the Options > Configure menu:

- Serial Port: Select /dev/serial0.
- Handshake: Set to None (unless you have wired the PROCEED/INTERRUPT lines).
- Baud Rate: Standard start is 19,200. Once verified, the Qt6 engine and Pi 4/5 hardware comfortably support high-speed divisors (up to 127kbps+).

**6. Telecommunications:**

Modem Bridge vs. R: Device Emulation AspeQt-2k26 features two distinct methods for connecting your Atari to Telnet BBSs and SSH hosts over the internet. It is important to understand the difference to choose the correct setup.

**Option A: The Legacy Modem Bridge (RS232 to TCP) – See Modem.pdf for more detail**

- How it works: This mode requires you to own physical Atari interface hardware, such as an original Atari 850 Interface module, a PR:Connection, or an R-Verter. The Raspberry Pi acts purely as a "dumb" Hayes modem. You connect a USB-to-Serial cable from the Pi to your Atari 850's RS232 port.

- Setup: In Options > Hayes Modem Emulator, check "Enable Modem Bridge". Select the USB serial port your 850 is plugged into (e.g., ttyUSB0) and set the baud rate to match your terminal program.

**Option B: The R: Device 850 Emulator (R: to TCP)**

- How it works: This mode entirely replaces the need for physical interface hardware. The Raspberry Pi intercepts the $40 (Poll) and $58 (Stream) SIO commands directly from the SIO bus. It loads a virtual 850 handler into the Atari's memory and acts as a fully integrated concurrent-mode modem.
- The Pi uses microsecond-level hardware interrupts to detect when the Atari drops the COMMAND line to exit the stream. (Note: Standard USB FTDI adapters can also achieve this emulation on modern PCs, but they require the CTS logic state to be explicitly inverted to mimic legacy RS-232 behavior—a hurdle bypassed entirely by wiring directly to the Pi's GPIO).
- Setup: 1. In Options > Hayes Modem Emulator, check "Enable 850 Device (R: to TCP)". 2. This will automatically enable "Direct Hardware UART (RasPi)" and force "CTS" handshaking to ensure the GPIO hardware interrupts are active. 3. You may select a phonebook.xml file to enable the built-in auto-dialer and macro injection (ESC-U for User, ESC-P for Pass).
- Terminal Setup (Ice-T / BobTerm): When booting your terminal software on the Atari, it is highly recommended to set your terminal baud rate to 9600 baud. The Pi 5 UART will automatically detect the speed change request from the Atari and drop down from the standard 19200 SIO speed to cleanly handle the 9600 baud stream.

**7. Troubleshooting & Performance**

Tuning Interfacing vintage 1980s hardware with a modern 64-bit Linux environment can sometimes present timing hurdles. If you encounter "Timeout" errors or "Checksum" failures, check the following:

**Common SIO Timing Issues**

- The "Slow-Speed" Fallback: If your Atari fails to boot the first time, ensure the baud rate is locked at 19,200. Some OS loaders require the initial handshake to be strictly at the standard speed before switching to high-speed routines.

- Linux Latency (CPU Scaling): By default, the Pi may throttle its CPU to save power, which can jitter the bit-banging timing of the UART.
    - Fix: Set your Pi to "Performance" mode by running: echo performance | sudo tee /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor
- The Level Shifter Bottleneck: Cheap bi-directional level shifters sometimes struggle with high-frequency signals. If you cannot reach speeds above 54kbps, ensure your shifter is rated for high-speed I2C/SPI (at least 2MHz).

**Handshaking and GPIO 14/15**

- Permissions: If AspeQt cannot open /dev/serial0, ensure your user is part of the dialout group: sudo usermod -a -G dialout $USER *(requires a log-out/log-in to take effect)*.

Document provided by 13leader.net. Visit the BBS via Telnet for community support and TNFS updates.